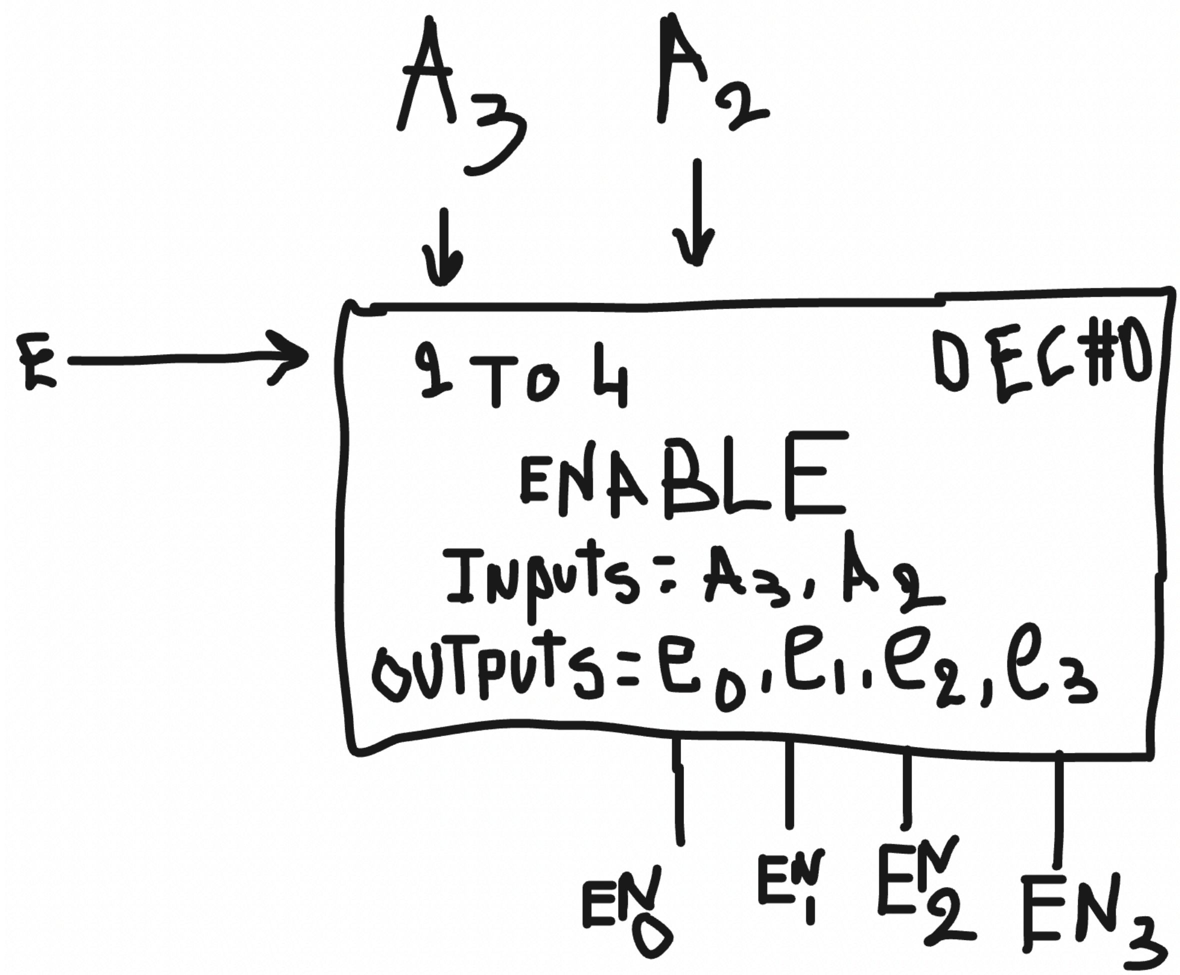
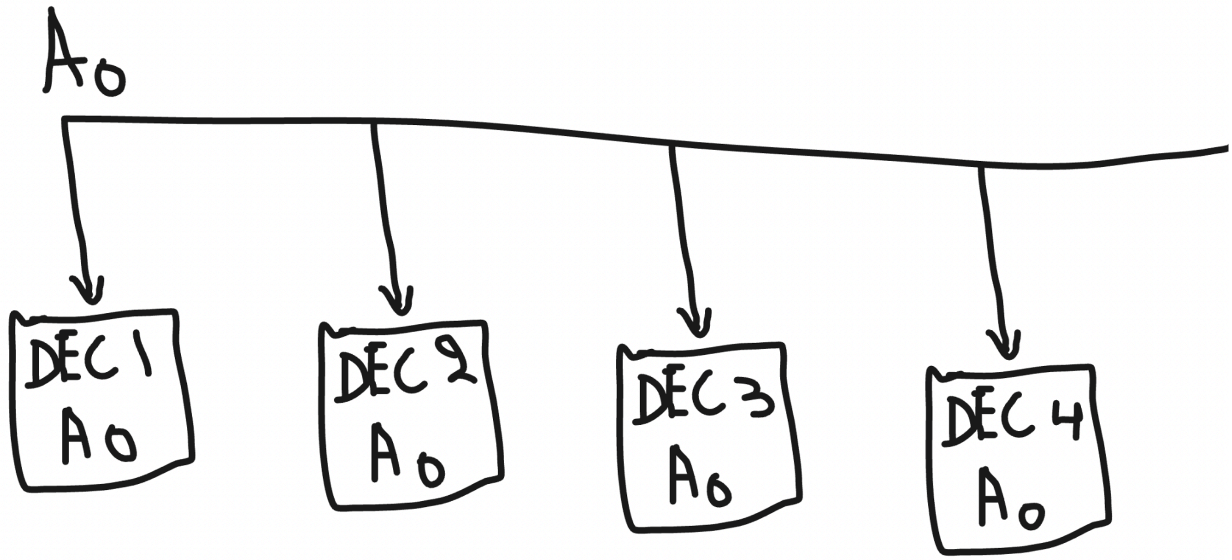
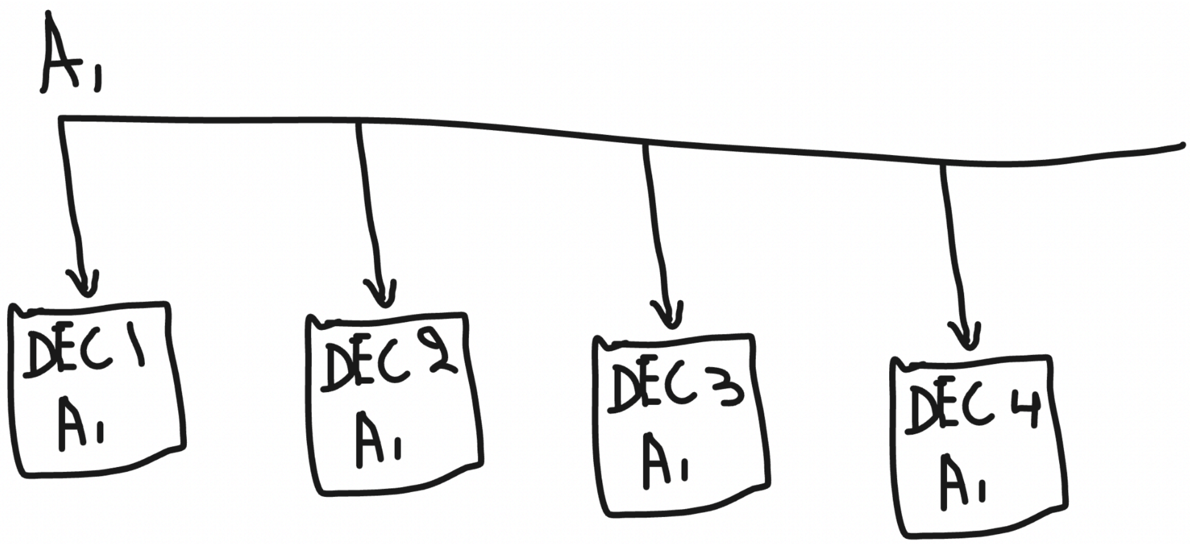
Harder Problem 1

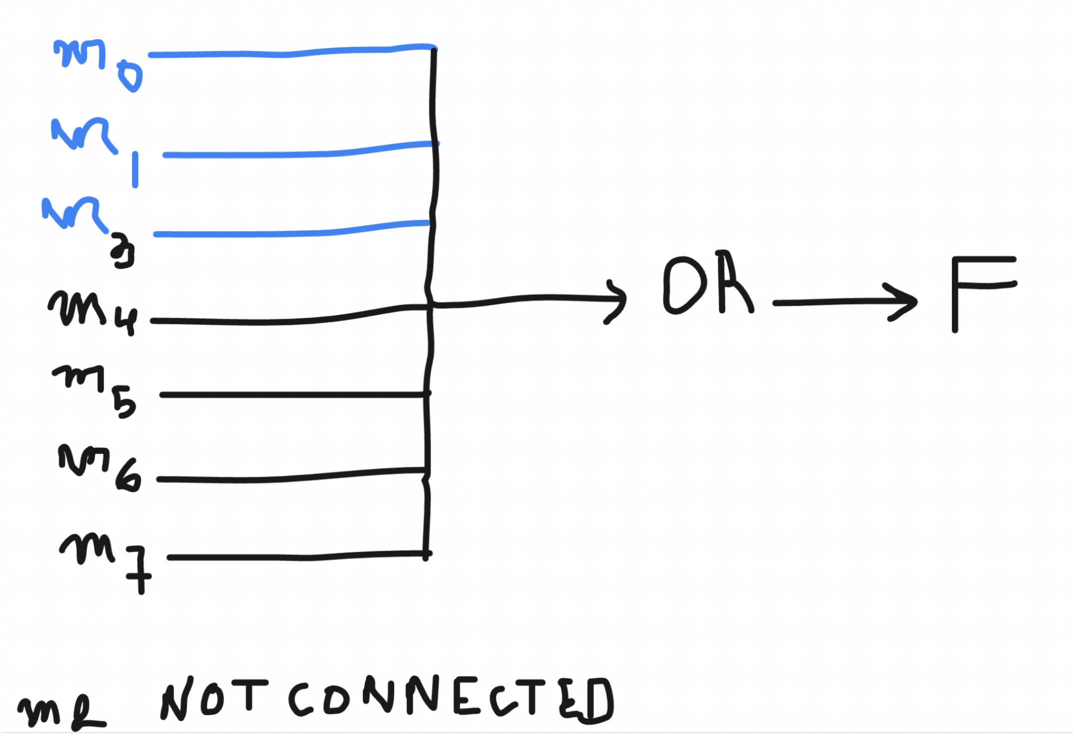



A_0
DEC 1
A_0
DEC 2
A_0
DEC 3
A_0
DEC 4
A_0


A_1
DEC 1
A_1
DEC 2
A_1
DEC 3
A_1
DEC 4
A_1

Harder Problem 2

F = m0 + m1 + m3 + m4 + m5 + m6 + m7



Harder Problem 3

so we assume A is the MSB and D is the LSB of ABCD.

NSL = A' B C D'

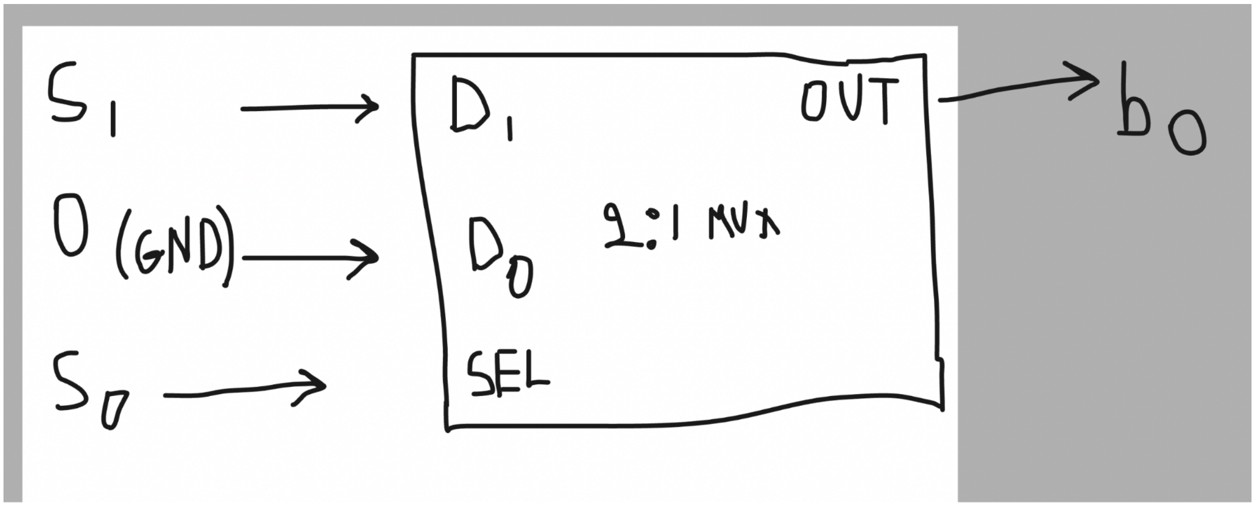
NSH = A + B C D

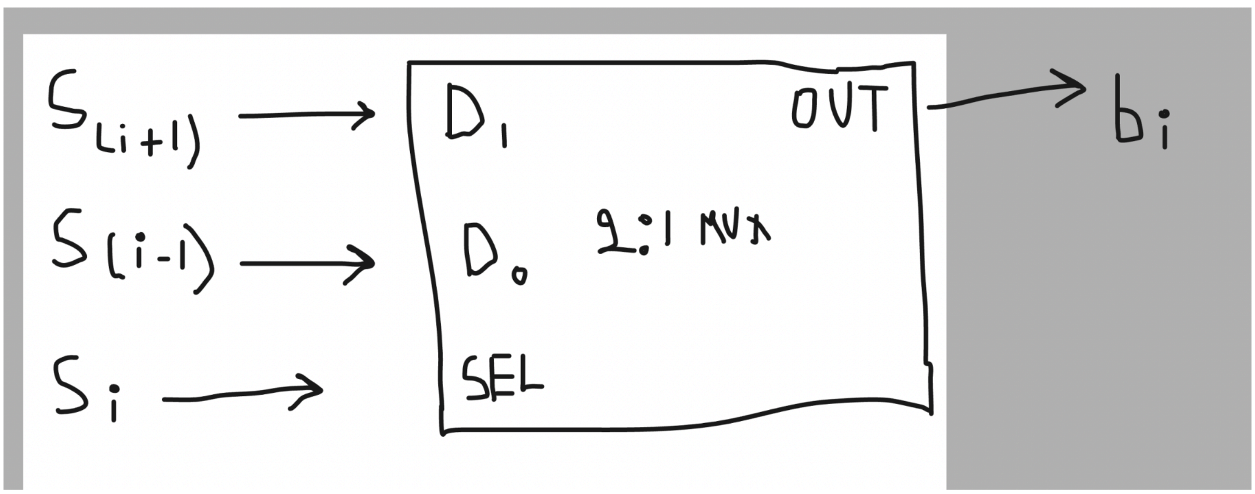
EWL = A B C D'

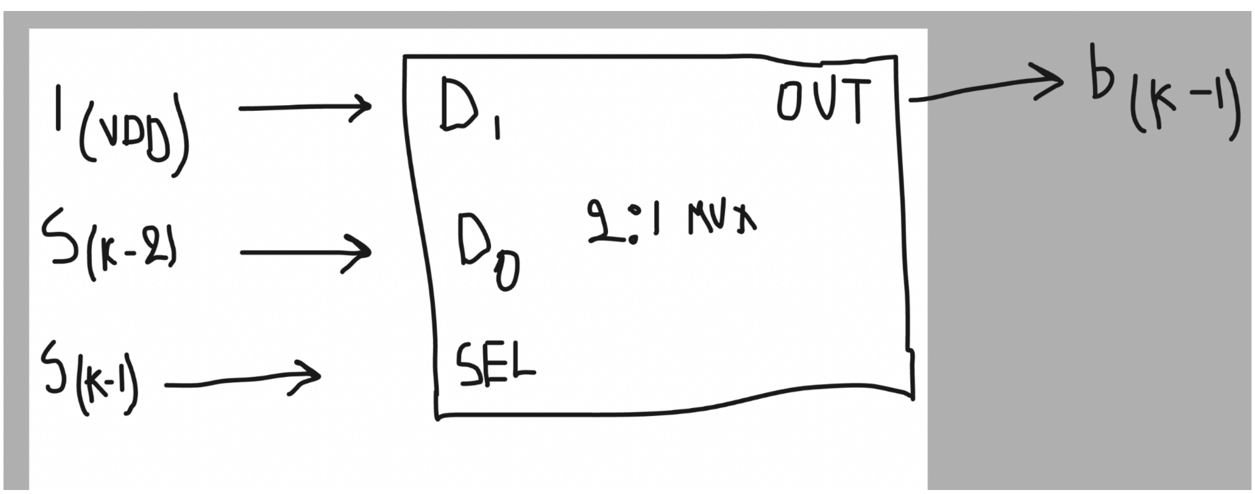
EWH = A' + B C D

+ = OR, adjacency = AND, and ' = NOT

Harder Problem 4







Harder Problem 5:

5(a):

so output bit name i'll call it T[i], T[i] = (S[i] AND S[i+1]) OR (S[i-1] AND NOT S[i-2]), so T[i] = (S[i] & S[i+1]) | (S[i-1] & ~S[i-2])

5(b): simplified equation: T[i] = (S[i] AND S[i+1]) OR (S[i-1] AND NOT S[i-2]) so T[i] = (S[i] & S[i+1]) | (S[i-1] & ~S[i-2])

5(c): simplified is T[i] = S[i] OR NOT S[i-2]

5(d): T[0] = S[0] AND S[1] so T[0] = S[0] & S[1]

the output bit T[1] is T[1] = S[0] OR (S[1] AND S[2])

output bit T[k-1] is T[k-1] = S[k-2] AND (S[k-1] OR NOT S[k-3])